Role Of Physiology in Craniofacial module

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- Physiology plays a foundational role in the **craniofacial module** for 1st year BDS because it explains how structures of the head and neck *function*, complementing anatomy and histology.
- **Physiology in the craniofacial module explains the functional mechanisms of muscles, glands, nerves, and tissues of the head and neck, forming the basis for understanding normal oral function and clinical dental practice.”**

Why It’s Important in BDS

- Helps you understand **normal function before studying disease**.
- Essential for:
  - Diagnosis of oral conditions
  - Understanding dental procedures (local anesthesia, extractions)
  - Clinical subjects like oral pathology and prosthodontics

1. Neuromuscular Function

- Physiology explains how craniofacial muscles (mastication, facial expression) contract and coordinate.
- Role of cranial nerves (especially V, VII, IX, X, XII) in:
  - Chewing
  - Swallowing
  - Speech
- Reflexes like the **jaw jerk reflex** and swallowing reflex are key concepts.

2. Mastication & Deglutition

- Mechanism of chewing: coordination between teeth, muscles, TMJ, and neural control.
- Saliva secretion and its role in lubrication and digestion.
- Phases of swallowing:
  - Oral phase (voluntary)
  - Pharyngeal phase (involuntary)
  - Esophageal phase

3. Salivary Gland Physiology

- Structure and function of major salivary glands (parotid, submandibular, sublingual).
- Regulation:
  - Parasympathetic → watery saliva
  - Sympathetic → viscous saliva
- Functions:
  - Digestion (amylase)
  - Protection (antibacterial action)
  - Buffering (pH control)

4. Special Senses in Craniofacial Region

- **Taste (gustation):** taste buds, neural pathways.
- **Smell (olfaction):** olfactory receptors and adaptation.
- **Vision & hearing basics** (relevant due to cranial nerve involvement).

5.Pain & Sensation

- Mechanism of orofacial pain:
  - Role of trigeminal nerve
- Types of pain:
  - Dental pain
  - Referred pain
- Basis for anesthesia in dental procedures.

6. Tooth Physiology

- Pulp function (nutrition, sensation, defense).
- Dentin sensitivity mechanism (hydrodynamic theory).
- Enamel and saliva interaction in remineralization.

7.Bone Growth & Development

- Physiology explains craniofacial growth patterns:
  - Intramembranous ossification (skull bones)
  - Endochondral ossification (base of skull)
- Hormonal influence (growth hormone, thyroid hormone).